NEXT GEN
SOFTWARE HUB

CYBER SECURITY INTERNSHIP

TASK 01 REPORT

Understanding the CIA Triad

Guarding the Digital World: Understanding the CIA Triad with Real-World Examples



TOPICS

- Confidentiality
- Integrity
- Availability

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1. Introduction to Cybersecurity

Cybersecurity is the practice of protecting information, computer systems, networks, applications, and other digital resources from unauthorized access, misuse, modification, disruption, or destruction. Organizations today depend heavily on information systems to perform their daily activities. Banks use computer systems to process financial transactions, businesses store customer information electronically, universities use digital platforms for teaching and administration, and government institutions use information systems to provide important services.

Because organizations depend on technology, cybersecurity is important for protecting both information and the systems that support business operations. A cybersecurity incident can result in the loss or exposure of sensitive information, unauthorized changes to data, disruption of services, financial losses, and damage to an organization's operations.

One of the fundamental concepts used to understand information security is the CIA Triad, which consists of Confidentiality, Integrity, and Availability. These three principles provide a foundation for understanding what needs to be protected in an information system. A single security incident may affect one or more of these principles.

For example, if an attacker steals customer information, confidentiality has been affected. If an attacker changes information without authorization, integrity has been affected. If an attack causes an important system to stop working, availability has been affected.

Understanding cybersecurity fundamentals therefore helps cybersecurity professionals identify threats, understand attacks, protect information systems, select appropriate security controls, and reduce the possible impact of security incidents.

2. Cybersecurity Fundamentals

Cybersecurity involves more than simply installing antivirus software or creating passwords. It is a broad area that involves protecting information, devices, networks, applications, users, and digital services against different types of security risks.

2.1 Information Security

Information security focuses on protecting information from unauthorized access, modification, disclosure, disruption, or destruction.

For example, a bank needs to protect:

- Customer personal information
- Account information
- Passwords and authentication credentials
- Financial transactions
- Internal business documents
- Security information

If this information is accessed or modified by unauthorized individuals, the organization and its customers may experience serious consequences.

2.2 Security of Computer Systems

Computer systems must be protected against malicious software, unauthorized access, vulnerabilities, and other attacks. Security measures may include access controls, authentication, security updates, antivirus software, system monitoring, and secure configuration.

2.3 Network Security

Networks connect computers, servers, applications, and users. Network security involves protecting network infrastructure and communications against unauthorized access and malicious activity.

Examples of network security controls include:

- Firewalls
- Network monitoring
- Intrusion detection and prevention
- Secure network configurations
- Access controls
- VPNs

2.4 Authentication and Access Control

Authentication is used to verify the identity of a user or system. For example, a username and password can be used to verify a user's identity.

Organizations can strengthen authentication through Multi-Factor Authentication (MFA), where users provide additional verification beyond a password.

Access control then determines what an authenticated user is allowed to access or modify. The principle of least privilege means users should only receive the access necessary to perform their responsibilities.

2.5 Security Awareness

Users are also an important part of cybersecurity. Employees and consumers may become targets of phishing, social engineering, malicious attachments, fake websites, or other attacks.

Security awareness helps users recognize suspicious activities and make safer decisions when using systems and handling information.

3. Threats and Attacks

Understanding the difference between a threat and an attack is an important cybersecurity fundamental.

3.1 Cybersecurity Threat

A threat is a potential danger that could exploit a weakness or vulnerability and cause harm to an information system, network, application, or organization.

A threat does not necessarily mean that an incident has already occurred. It represents the possibility that something harmful could happen.

Examples of cybersecurity threats include:

- Malware
- Phishing
- Ransomware
- Unauthorized access
- Insider threats
- Denial-of-Service attacks
- Exploitation of software vulnerabilities

3.2 Cybersecurity Attack

A cybersecurity attack occurs when a threat is carried out against a computer system, network, application, account, or information resource.

For example, phishing is a threat when there is a possibility that users could be tricked into revealing their credentials. When an attacker actually sends a fraudulent message and successfully obtains a user's credentials, an attack has taken place.

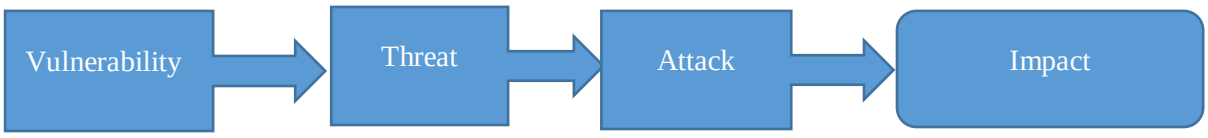
An attack can attempt to:

- Steal information
- Modify information
- Gain unauthorized access
- Install malicious software
- Disrupt services
- Destroy or encrypt data
- Make systems unavailable

3.3 Relationship Between Threats, Vulnerabilities and Attacks

A vulnerability is a weakness that can potentially be exploited. A threat represents the possibility of harm, while an attack is the action taken to exploit a weakness.

For example:



An organization may have an outdated application containing a security vulnerability. An attacker may identify this weakness as an opportunity. The attacker can then exploit the vulnerability through an attack. The resulting impact could include unauthorized access, data exposure, modification of information, or system disruption.

This demonstrates why organizations need to identify vulnerabilities and implement security controls before attackers can exploit them.

4. Common Cyber Threats

The Week 1 internship task specifically requires identifying different types of cyber threats, including Malware, Phishing, and Ransomware.

4.1 Malware

Malware is malicious software designed to perform unauthorized or harmful activities on a computer system, device, network, or other digital resource.

Malware can be used to:

- Steal information
- Damage files
- Monitor users
- Gain unauthorized access
- Disrupt systems
- Install additional malicious software
- Provide attackers with unauthorized control

Examples of malware include viruses, worms, Trojans, spyware, and ransomware.

How Malware Can Affect an Organization

Malware can affect all three areas of the CIA Triad.

Confidentiality: Malware may steal passwords, personal information, or confidential business data.

Integrity: Malware may modify, corrupt, or delete files and system configurations.

Availability: Malware may damage systems or prevent users from accessing important resources.

Malware Protection

Organizations can reduce malware risks by using:

- Antivirus and endpoint protection
- Security updates and patches
- Access controls
- Email security
- Regular backups
- Security monitoring
- User security awareness
- Safe software installation practices

4.2 Phishing

Phishing is a social engineering technique in which an attacker attempts to deceive a user into providing sensitive information or performing a harmful action.

Attackers may create messages that appear to come from legitimate organizations, banks, companies, universities, or other trusted sources.

A phishing message may attempt to convince a user to:

- Enter a username and password
- Provide financial information
- Open a malicious attachment
- Click a malicious link
- Download malicious software
- Provide authentication codes

For example, a user may receive an email claiming that their account has a security problem and asking them to click a link and log in. The link may lead to a fraudulent website designed to collect the user's credentials.

Impact of Phishing

Phishing can result in:

- Stolen usernames and passwords
- Unauthorized account access
- Financial fraud
- Malware infection
- Exposure of confidential information
- Further attacks against an organization
- Protection Against Phishing

Users and organizations can reduce phishing risks by:

- Checking the sender of messages
- Verifying suspicious requests
- Avoiding unknown links and attachments
- Using MFA
- Checking website addresses before entering credentials
- Providing security awareness training
- Using email security controls

4.3 Ransomware

Ransomware is a type of malicious software that typically encrypts or locks data and demands payment from the victim.

A ransomware attack can prevent an organization from accessing important files and systems. This can cause serious operational disruption, especially when the affected organization depends on digital systems for its daily activities.

Possible Effects of Ransomware

Ransomware can affect:

- File availability
- Business operations
- Customer services
- Financial operations
- Critical systems
- Data recovery processes

The most direct connection to the CIA Triad is Availability, because users may be unable to access systems or information. Depending on the incident, confidentiality and integrity can also be affected.

Ransomware Protection

Organizations can reduce ransomware risks by using:

- Regular and tested backups
- Endpoint security
- Security updates
- Access controls
- Least privilege
- Network segmentation
- Email security
- Security awareness training
- System monitoring
- Incident response procedures

Backups are particularly important because they can provide copies of information that may be used for recovery when original data becomes unavailable.

5. Security Controls

Security controls are measures used by organizations to reduce cybersecurity risks and protect information systems.

The controls selected should address the specific risks faced by an organization. Different controls can support confidentiality, integrity, availability, or more than one of these principles.

5.1 Confidentiality Controls

Confidentiality controls help prevent unauthorized access or disclosure of information.

Examples include:

Encryption: protects information by converting it into a protected format.

MFA: provides additional verification beyond a password.

Access control: determines who can access specific systems or information.

Strong passwords: reduce the risk of unauthorized account access.

Least privilege: limits users to only the access they need.

The 2017 Equifax data breach demonstrates why confidentiality controls are important. Attackers gained unauthorized access to sensitive personal information, including names, addresses, dates of birth, and Social Security numbers.

5.2 Integrity Controls

Integrity controls help prevent unauthorized modification and help organizations identify when information has been changed.

Examples include:

- Cryptographic hashing
- Digital signatures
- Access controls
- Audit logs

For example, hashing can help detect whether a file has changed, while audit logs can record which account performed an activity and when the activity occurred.

The SolarWinds supply-chain attack demonstrates the importance of integrity. Malicious code was inserted into legitimate software updates, meaning customers could receive compromised software through a trusted software provider.

5.3 Availability Controls

Availability controls help ensure that authorized users can access systems and information when they need them.

Examples include:

- Backups
- Redundant systems
- Disaster recovery plans
- DDoS protection
- System monitoring
- Incident response
- Business continuity procedures

These controls help organizations continue operations or recover services following a security incident or system failure.

6. Relationship Between Cyber Threats and the CIA Triad

Cyber threats and attacks can affect one or more components of the CIA Triad.

6.1 Threats Affecting Confidentiality

Confidentiality is affected when unauthorized individuals gain access to information.

For example, phishing may allow an attacker to steal a user's credentials. The attacker could then use those credentials to access confidential information.

Malware can also collect information without the user's knowledge.

The main security question for confidentiality is:

Who should be allowed to access this information?

6.2 Threats Affecting Integrity

Integrity is affected when information, software, configurations, or other resources are changed without authorization.

For example, an attacker who gains access to a database could modify customer information or financial records.

The SolarWinds incident demonstrates that integrity must also be protected throughout software development and distribution, not only after software is installed.

The main security question for integrity is:

How can we ensure that information remains accurate and has not been changed without authorization?

6.3 Threats Affecting Availability

Availability is affected when authorized users cannot access systems, services, or information when they need them.

Ransomware can make files unavailable by encrypting them. DDoS attacks can make online services unavailable by overwhelming them with malicious traffic.

The December 2023 Central Bank of Lesotho cybersecurity incident provides a local example. Some systems were suspended to prevent further infiltration, and the incident affected the National Payments System, making normal inter-bank transactions impossible during the outage.

The available report does not specify the exact technical nature of that attack, so it should not be classified as ransomware, DDoS, or another specific attack without additional evidence.

6.4 One Threat Can Affect Multiple CIA Principles

The CIA principles should not be treated as completely separate.

For example, consider a ransomware attack:

The attacker may gain unauthorized access to systems.

Confidential information may be exposed.

Files may be modified or encrypted.

Users may lose access to important systems and information.

Therefore, one incident can potentially affect:

Confidentiality

Integrity

Availability

This is why organizations need a balanced cybersecurity strategy that protects all three principles.

7. Key Lessons Learned

From this task, several important cybersecurity lessons can be identified.

7.1 Cybersecurity Is More Than Preventing Unauthorized Access

Cybersecurity is not only about stopping attackers from entering a system. Organizations must also ensure that information remains accurate and that important systems remain available.

The CIA Triad provides a framework for considering all three requirements.

7.2 Organizations Must Understand Different Threats

Malware, phishing, and ransomware operate differently and can produce different impacts. Understanding how these threats work helps organizations select appropriate security controls.

7.3 Human Users Are Important to Security

Users can become targets of phishing and social engineering attacks. Security awareness, strong authentication, and careful handling of emails and links can reduce the risk of successful attacks.

7.4 Security Controls Should Work Together

No single security control can provide complete protection. Organizations should combine controls such as authentication, encryption, access control, monitoring, backups, and incident response.

7.5 Third-Party Software Must Also Be Trusted Carefully

The SolarWinds example demonstrates that an organization can be affected through a trusted supplier. Organizations therefore need to consider the security and integrity of third-party software and services.

7.6 Recovery Is an Important Part of Cybersecurity

The Central Bank of Lesotho example demonstrates the importance of maintaining essential services during a cybersecurity incident. Organizations need backup systems, disaster recovery, monitoring, incident response, business continuity, and recovery capabilities.

7.7 The CIA Triad Helps Security Professionals Think About Impact

When analyzing a threat or attack, cybersecurity professionals can ask:

Confidentiality: Could unauthorized users access the information?

Integrity: Could the information or system be changed without authorization?

Availability: Could authorized users lose access to the system or information?

These questions help security professionals understand the possible impact of an incident and select appropriate controls.

8. Conclusion

Cybersecurity is an essential part of protecting modern information systems, networks, applications, and digital resources. Organizations depend on technology for financial transactions, communication, business operations, education, government services, and other important activities. This dependence also creates security risks that organizations must manage.

This task examined important cybersecurity fundamentals, including threats, attacks, malware, phishing, and ransomware. It also demonstrated how these threats can affect information systems and organizations.

The CIA Triad (Confidentiality, Integrity, and Availability): provides a fundamental framework for understanding cybersecurity requirements. Confidentiality ensures that information is protected from unauthorized access and disclosure. Integrity ensures that information remains accurate, complete, trustworthy, and protected from unauthorized modification. Availability ensures that authorized users can access systems, applications, services, and information when required.

The Equifax data breach demonstrates the importance of protecting confidentiality. The SolarWinds supply-chain attack demonstrates the importance of protecting integrity throughout software development and distribution. The Central Bank of Lesotho incident demonstrates the importance of maintaining availability for critical financial services.

Overall, effective cybersecurity requires organizations to understand threats, identify vulnerabilities, implement appropriate security controls, and consider the potential effect of incidents on confidentiality, integrity, and availability. A secure information system should keep information private when it needs to be private, trustworthy when it needs to be trusted, and available when it needs to be available.